various states of the air molecules can be described by giving the state of the air molecules in the first container, and then by giving the state of those in the second. Thus, the number of arrangements for the joined containers is the square of the number of arrangements of either separately. After taking the logarithm, this tells us that the entropy has doubled.

2. You will note that it doesn't really make much sense to compare a volume with an area, as they have different units. What I really mean here, as indicated by the text, is that the rate at which volume grows with radius is much faster than the rate at which surface area grows. Thus, since entropy is proportional to surface area and not volume, it grows more slowly with the size of a region than it would were it proportional to volume.

3. While this captures the spirit of the entropy bound, the expert reader will recognize that I am simplifying. The more precise bound, as proposed by Raphael Bousso, states that the entropy flux through a null hypersurface (with everywhere non-positive focusing parameter Θ) is bounded by A/4, where A is the area of a spacelike cross-section of the null hypersurface (the "light-sheet").

4. More precisely, the entropy of a black hole is the area of its event horizon, expressed in Planck units, divided by 4, and multiplied by Boltzmann's constant.

5. The mathematically inclined reader may recall from the endnotes to Chapter 8 that there is another notion of horizon—a cosmic horizon—which is the dividing surface between those things with which an observer can and cannot be in causal contact. Such horizons are also believed to support entropy, again proportional to their surface area.

6. In 1971, the Hungarian-born physicist Dennis Gabor was awarded the Nobel Prize for the discovery of something called *holography.* Initially motivated by the goal of improving the resolving power of electron microscopes, Gabor worked in the 1940s on finding ways to capture more of the information encoded in the light waves that bounce off an object. A camera, for example, records the intensity of such light waves; places where the intensity is high yield brighter regions of the